

Playbook: Suspicious PowerShell Flags (EQL)

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MITRE ATT&CK: T1059.001 (PowerShell) — Tactic: Execution

Purpose

Detect potentially obfuscated or headless PowerShell executions commonly used by attackers (e.g., EncodedCommand, NoProfile, -nop). This rule is validated in Elastic and produces alerts with end-to-end evidence for interviews and Capstone.

Data Sources

- Elastic Defend (endpoint) process telemetry
- Indices: `logs-\*`, `endpoint.events.\*`

Detection (EQL)

```
process
  where process.name : ("powershell.exe", "pwsh.exe")
  and (
    process.command_line : "-enc*" or
    process.command_line : "-EncodedCommand*" or
    process.command_line : "-NoProfile*" or
    process.command_line : "-nop*"
  )
```

Rule Configuration (Elastic)

- **Type:** EQL — Event category: `process`
- **Index patterns:** `logs-\*`, `endpoint.events.\*`
- **Schedule:** Every 5 minutes
- **Additional look-back:** 1 minute
- **Severity:** High
- **Risk score:** 75
- **Tags (optional):** `["MITRE:T1059.001", "PowerShell", "Execution"]`

Validation Steps (What was done)

1. Verified incoming telemetry in *Discover* (`process.name:"powershell.exe" and process.command\_line:"EncodedCommand"`).
2. Manually triggered test events:
 - `powershell.exe -NoProfile -EncodedCommand VwByAGkAdABlAC0ATwB1AHQAcAB1AHQAIB0AGUAcwB0AA==`
 - `powershell.exe -NoProfile -nop -Command "Write-Output test"`
3. Confirmed alerts in *Security* → *Alerts* filtering on `kibana.alert.rule.name: "Suspicious PowerShell Flags (EQL)"`.

Evidence (Artifacts)

- `12\_rule\_manual\_run.png` — Manual run succeeded for the rule
- `13\_agent\_reenrolled\_healthy.png` — Agent healthy after re-enroll
- `14\_discover\_verified\_powershell.png` — Telemetry shows PowerShell process events
- `15\_discover\_cmdline\_flags.png` — Command-line flags present
- `16\_rule\_manual\_run\_success.png` — Preview shows matches (or successful execution)
- `17\_alerts\_list\_powershell\_eql.png` — Alert list with hits
- `18\_alert\_details\_powershell\_eql.png` — Alert details include command line

Response

- Triage the command, parent/child chain, and user/session context.
- Contain host if malicious intent is confirmed (disable network, isolate in EDR).
- Hunt for similar executions over the last 24–72 hours; pivot on user, host, and `process.parent.\*` fields.

Reporting

- Record the incident summary (time, user, host, command line, outcome).
- Map to **MITRE T1059.001** in the incident write-up and Capstone Task 2 framework alignment.

Notes

- Use `kibana.alert.rule.name` (not `rule.name`) for filtering in Alerts in Elastic 8/9.x.